

Qiming Liu

liuqm110@gmail.com | (417) 234-8164 | Springfield, MO | [Linkedin.com/in/qiming-liu-845ba92a0/](https://www.linkedin.com/in/qiming-liu-845ba92a0/)

EDUCATION

Bachelor of Science in Data Analytics

Expected Graduation: May 2027

Missouri State University | Springfield, MO | GPA: 3.94/4.0

- Relevant Coursework: Database Management, Applied Business Statistics, Data Mining, Machine Learning

EXPERIENCE

Intern — AI Engineer

August 2026 — Present

OriLoan Financial Inc.

- Engineered an automated pricing-ingestion and reconciliation platform covering 5 mortgage lenders across 5 heterogeneous portal architectures (REST API, replayable HTTP, ASP.NET WebForms, Blazor/SignalR, JSF), replacing manual rate-sheet processing and validating every quoted rate against each lender's official pricing engine.
- Reverse-engineered undocumented off-sheet pricing (service-release premium/margin) into a deterministic model, raising pricing accuracy from 44.6% to 99.8% against lender ground truth (out-of-sample, cross-validated over 19,498 rate points), resolving the core blocker to a launch-ready consumer rate product.
- Implemented scenario-level k-fold cross-validation and gated all 5 lenders in automated CI regression, ensuring pricing accuracy is continuously verified and protected from production regressions.
- Launched and hardened a public pricing API (per-caller rate limiting, input validation, CloudWatch observability), and accelerated delivery by directing AI coding agents to build importers, run large-scale pricing sweeps, and analyze residuals.

Research

Jun 2026 – Present

Medicaid Expansion and County-Level Health & Economic Outcomes in Missouri — Spatial Difference-in-Differences

- Investigating whether Missouri's 2021 Medicaid expansion causally improved county-level health and economic outcomes, and whether those gains spill over to neighboring counties, producing the county-level evidence that prior state- and national-level studies lack, to inform state healthcare policy.
- Assembled a panel dataset spanning all 114 Missouri counties from five federal sources, engineering health and economic indicators: uninsured rate, preventable hospitalizations, diabetes mortality, poverty, median income, education, and age structure.
- Specifying a Spatial Difference-in-Differences model with county and year fixed effects to isolate the treatment effect and quantify cross-county spillovers across the state.

PROJECTS & CERTIFICATIONS

Loan Default Prediction — America's Innovate IT Collegiate Conference (AITCC) Analyze IT Challenge | 2nd Place

March 2026

- Joined and cleaned three relational tables (70K borrowers, 180K loans, 627K payment records) in Python, handling missing values, duplicates, orphaned foreign keys, and inconsistent formatting to produce a single loan-level modeling dataset.
- Engineered payment behavior features from monthly repayment records including on-time rate and cumulative delinquent days. Correlation analysis showed these far outperformed traditional metrics like credit score and income ($r = 0.76$ vs near-zero).
- Compared Logistic Regression, Random Forest, and XGBoost with stratified validation. Best models achieved AUC 0.96 (classification) and R^2 0.79 (regression). Extended the pipeline into an operational risk monitor with monthly KPI tracking, risk segmentation, a Tableau dashboard, and AI-generated executive reports via Claude API.

Automated Daily Operations Monitoring Pipeline

- Built an automated monitoring pipeline to convert raw operational data into daily KPIs and flag anomalies using a rolling 7 day baseline, improving visibility into daily operational performance.
- Orchestrated scheduled runs via n8n to automate KPI reporting into Google Sheets, achieved 100% automation of the data pipeline from raw ingestion to visualization, eliminating the need for manual data entry.
- Improved monitoring consistency by standardizing daily checks and enabling earlier detection of revenue drops and cancellation spikes, supporting faster and more proactive operational decision making.

IBM Data Analyst Professional Certificate

- Executed end to end analytical workflows using Python, SQL, Excel, AI assisted analysis, and data visualization tools to cleanse, interrogate, and synthesize real world datasets into dashboards and decision ready insights.

SKILLS

- Programming & Tools: Python, R, SQL, Git, pandas, scikit-learn, Advanced Excel, Tableau, Power BI
- Machine Learning & Statistics: Random Forest, XGBoost, Cross-Validation, Feature Engineering, EDA, Difference-in-Differences
- Engineering & Automation: ETL, REST APIs, CI/CD, AI-Assisted Development